Predicting Type II Diabetes Risk Using Explainable Machine Learning

A Healthcare AI Approach for Early Risk Identification using BRFSS 2023 Data

Presented by: Apex

Apex AI

Why Diabetes? The Global Challenge

🩺 589M
Adults living with diabetes worldwide

🌍 252M
Adults remain undiagnosed globally

🇰🇪 813,300
Adults living with diabetes in Kenya

📈 3.1%
Adult diabetes prevalence in Kenya

Source: IDF Diabetes Atlas, 2021

The Clinical Problem: Late Diagnosis Impact

Undiagnosed Patient
Millions unaware of their risk

Delayed Detection
Symptoms appear years too late

Complications & Hospitalization
Neuropathy, retinopathy, organ damage

Can AI predict risk earlier?

Project Objectives

Explore demographic, behavioral, and clinical factors associated with diabetes.

Develop a predictive Machine Learning Model.

Identify the most influential predictors and translate them into actionable recommendations.

Implement SHAP and LIME for model explainability

develop an interactive dashboard for disease risk prediction and visualization

Primary Stakeholders:
Clinicians and healthcare providers
Heath promotion programs
Individuals at risk of T2DM

Secondary Stakeholders:
Research and Academic institutions
Health Insurance Organizations
Policy makers
NGOs and Community Health Organizations

Dataset Overview: CDC BRFSS 2023

CDC Behavioral Risk Factor Surveillance System: 433,323 participants surveyed → 350 variables collected → 429,086 final records after cleaning. The largest telephone health survey system in the world.

Data Source

CDC BRFSS 2023 Behavioral Risk Factor Surveillance Survey.

Data Info

433,323 participants 350 variables collected

Final Data

429,086 final records retained after preprocessing.

Features Selected

15 key features selected

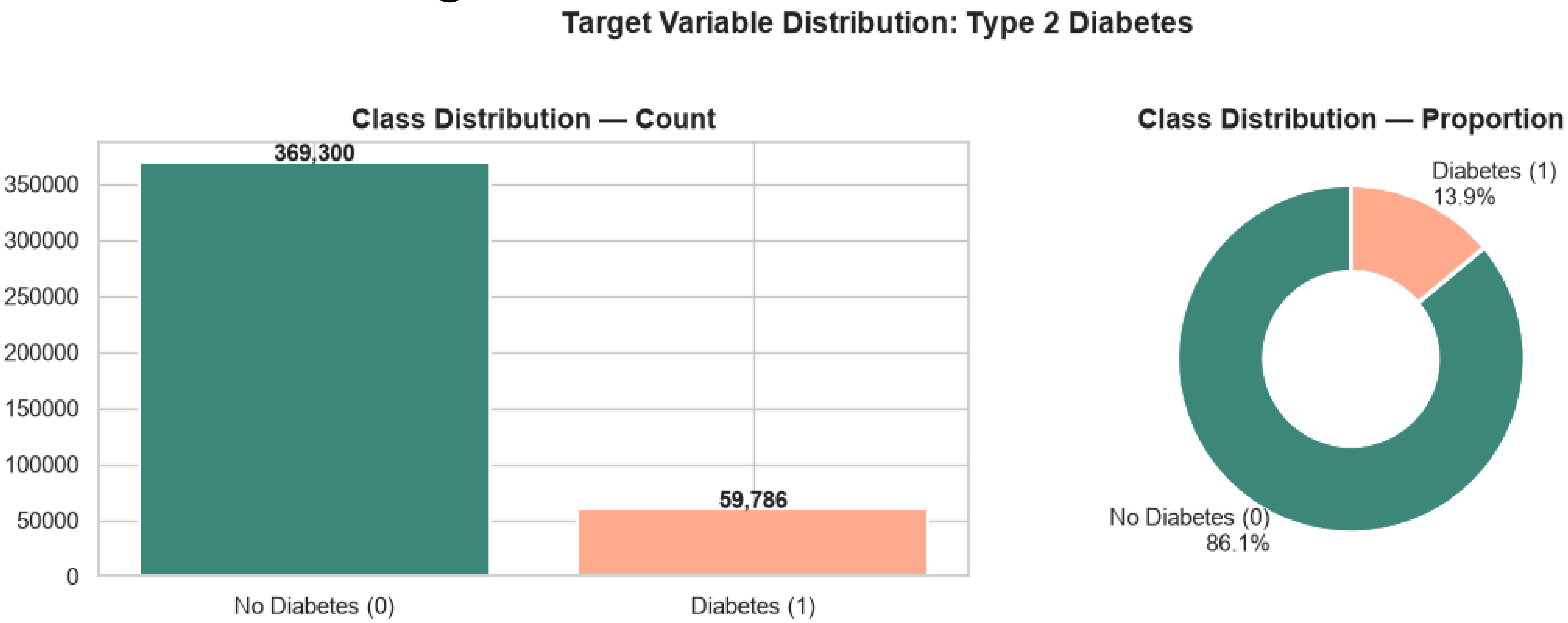
Target variable

Type II Diabetes diagnosis (binary).

Data Source CDC / BRFSS 2023

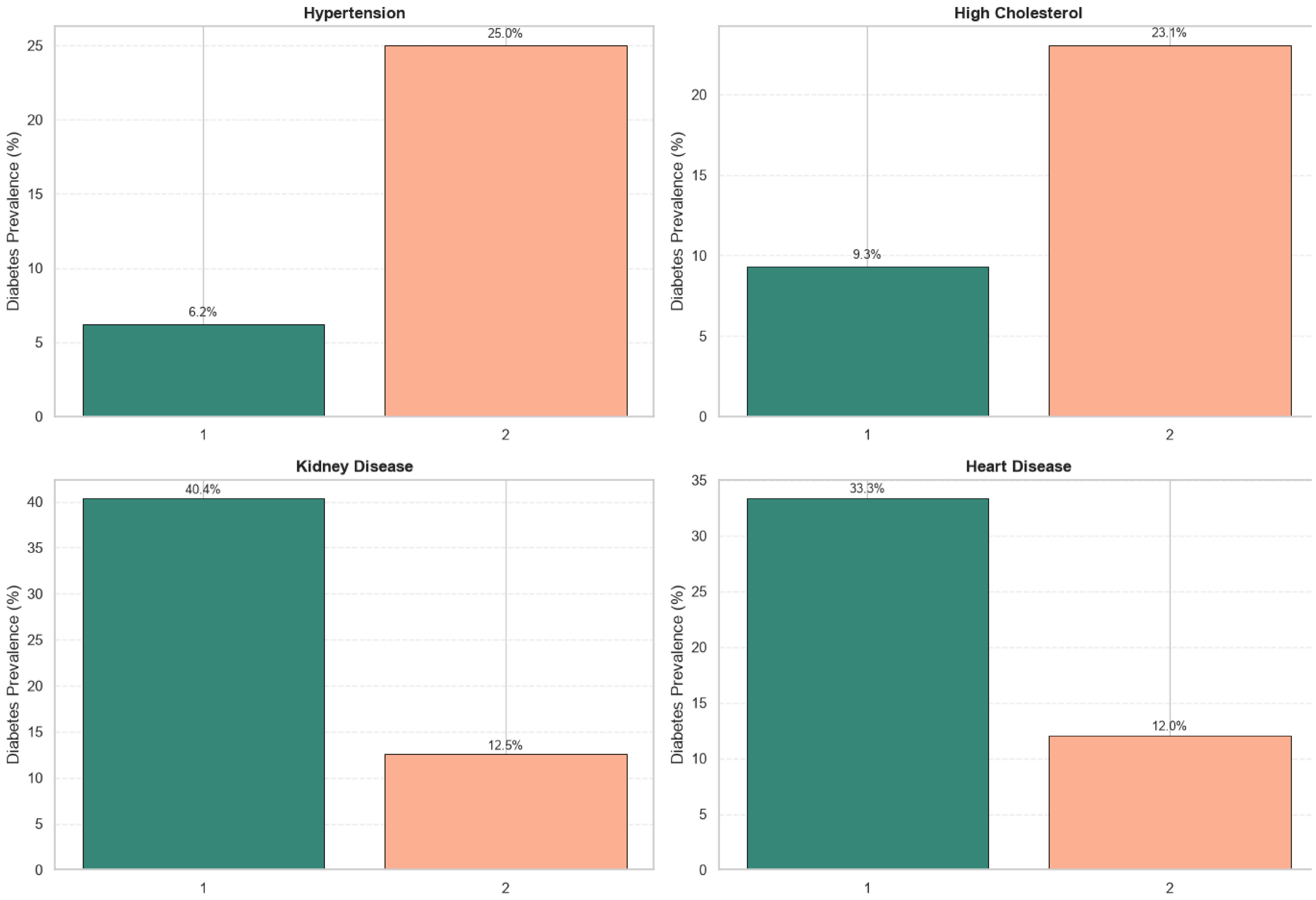
CDC 433323

Target Variable Distribution



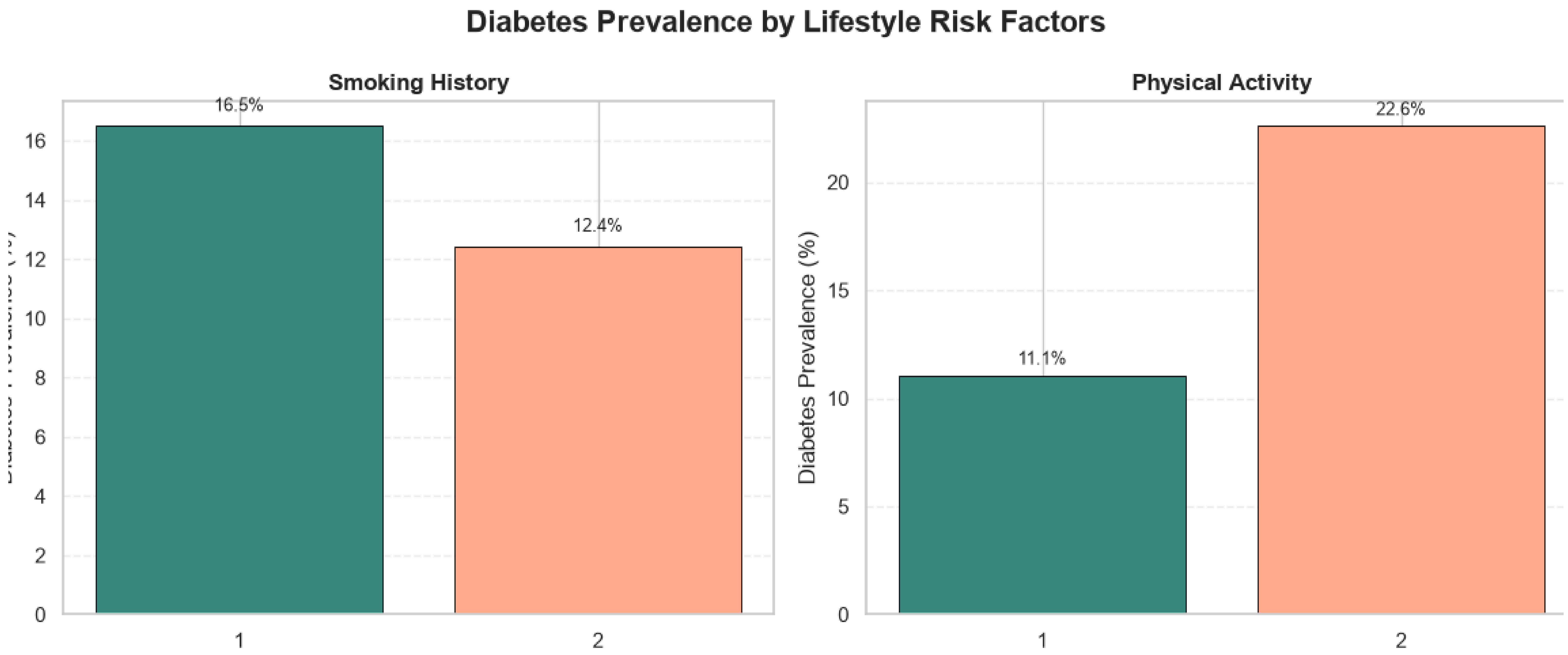
- Class imbalance: 85.7% non-diabetic vs. 14.3% diabetic, consistent with real-world prevalence.
- Evaluation focus: **Recall** was prioritized over accuracy to reduce missed diabetes cases (false negatives).

CLINICAL RISK FACTORS



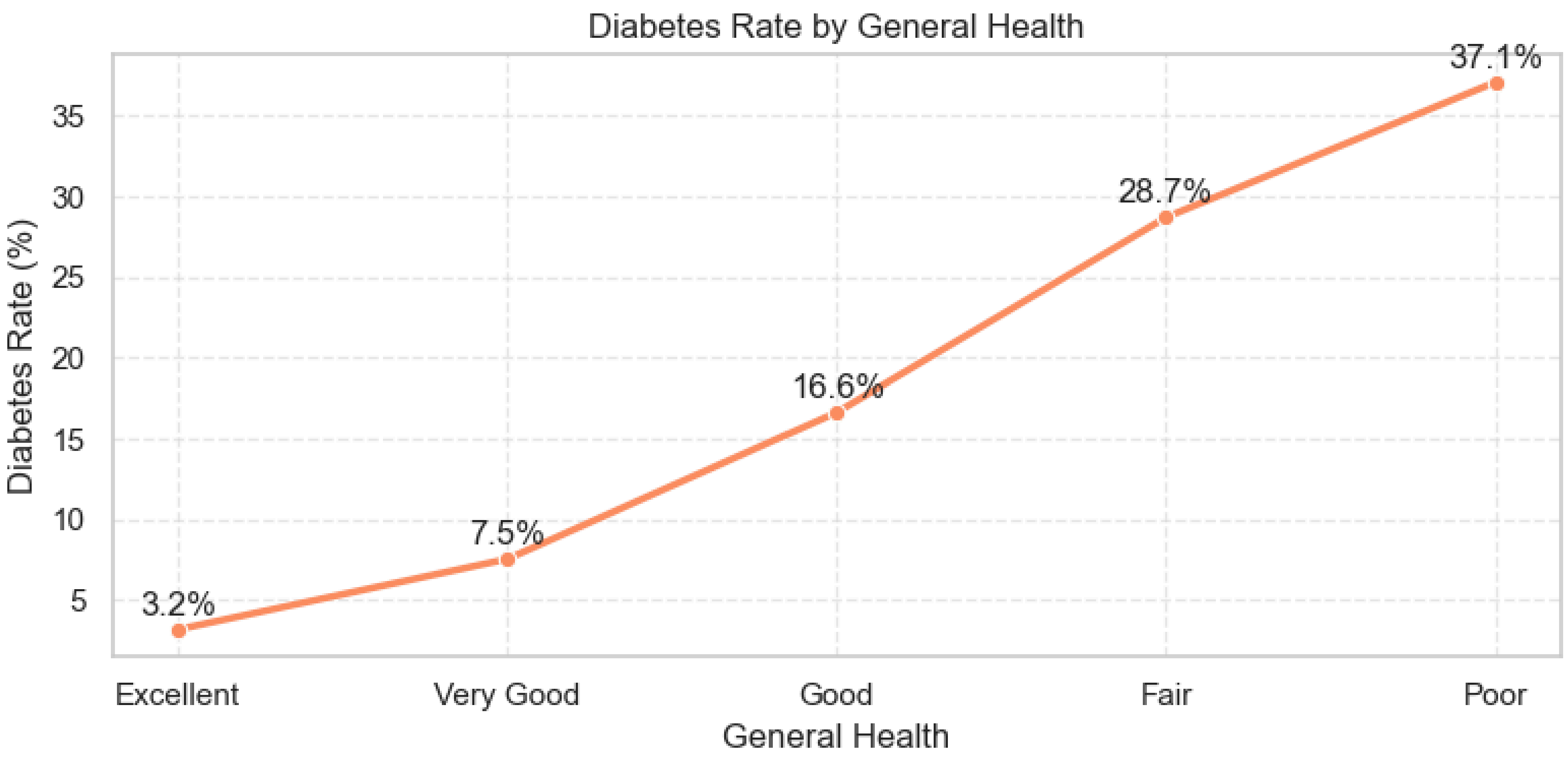
- Diabetes prevalence was consistently higher among individuals with hypertension, high cholesterol, heart disease, and kidney disease.
- Kidney disease and heart disease showed the strongest association with diabetes.
- These findings align with clinical evidence, supporting their inclusion as key predictive features.

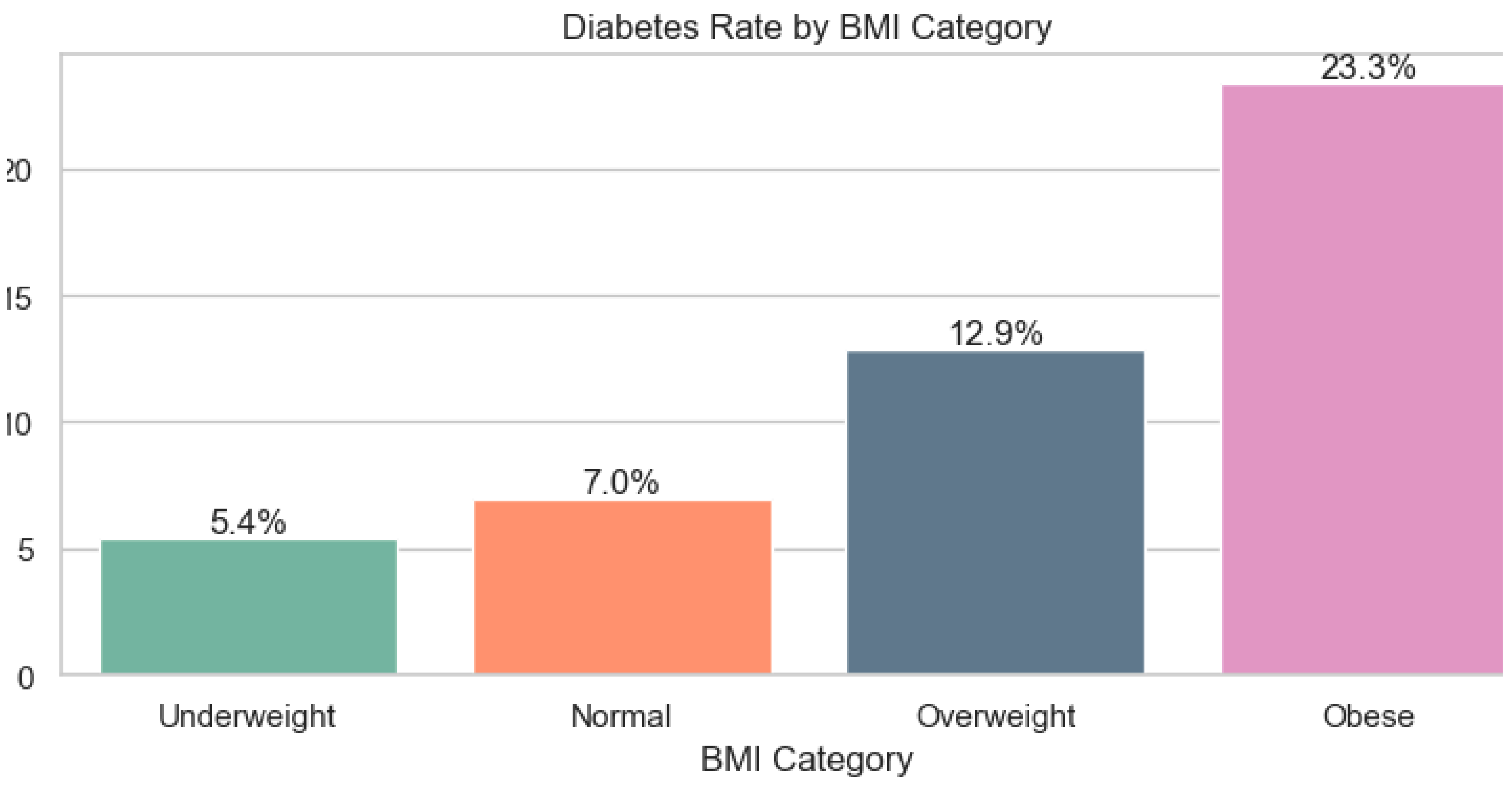
LIFESTYLE RISK FACTORS



- Higher diabetes prevalence among physically inactive individuals.
- Smoking was associated with increased diabetes risk
- Supports lifestyle behaviors as important predictive features.

Diabetes Prevalence by General Health and BMI

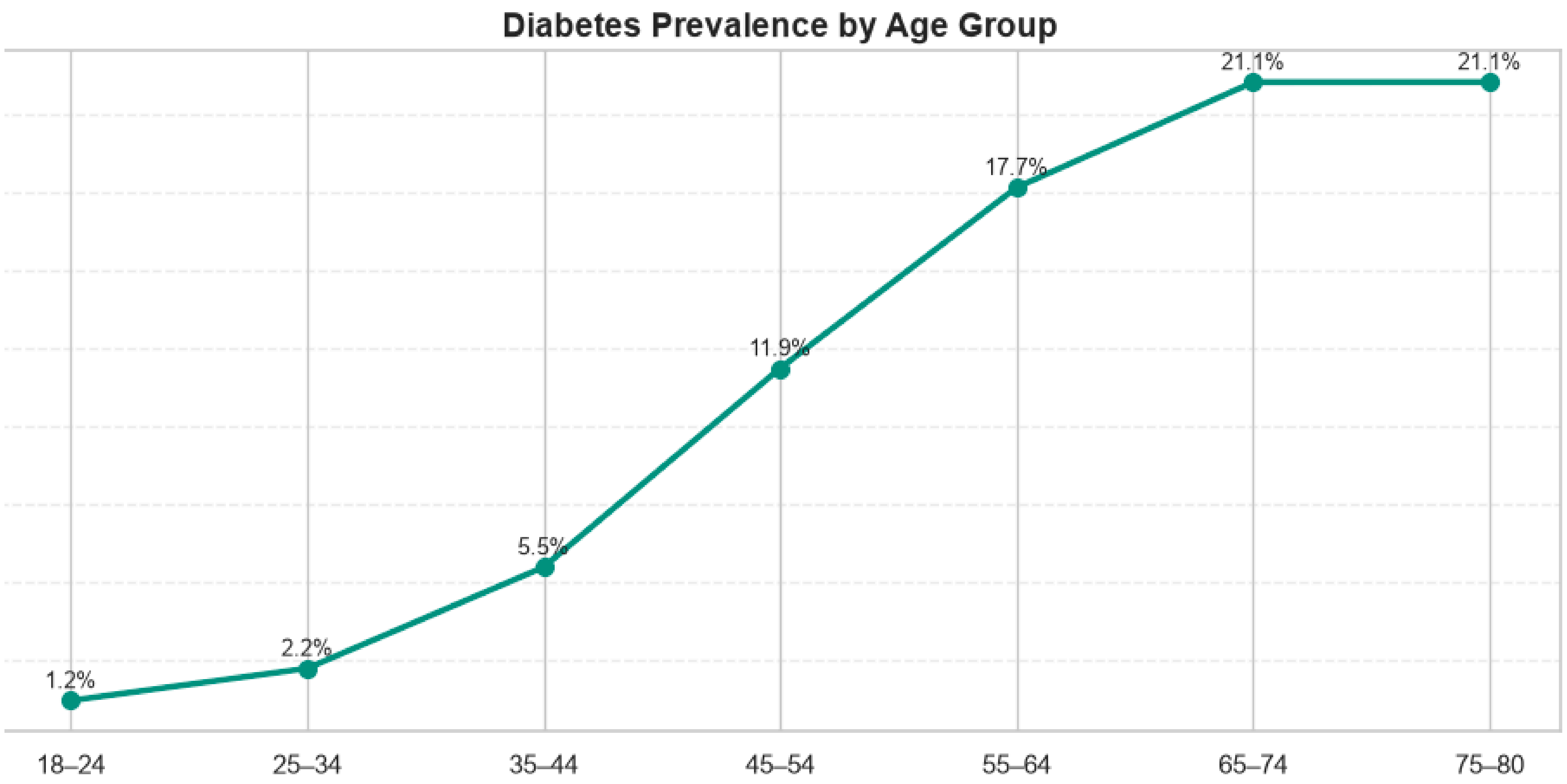




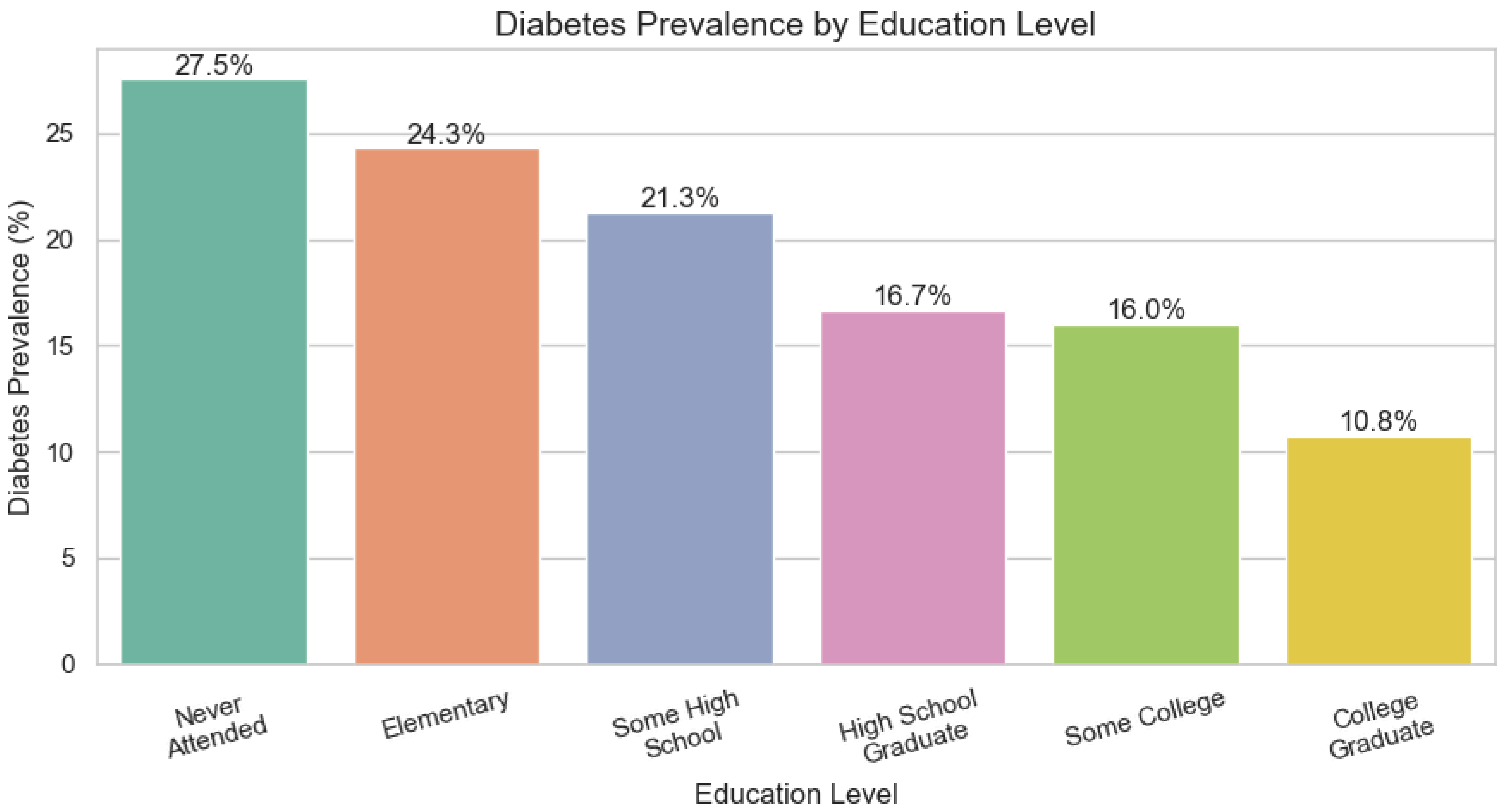
- The steady upward trend indicates a strong association between poorer general health and Type 2 Diabetes.

- The chart shows a clear upward trend between BMI category and diabetes rate. Diabetes prevalence increases from 5.4% among underweight respondents to 23.3% among respondents with obesity.

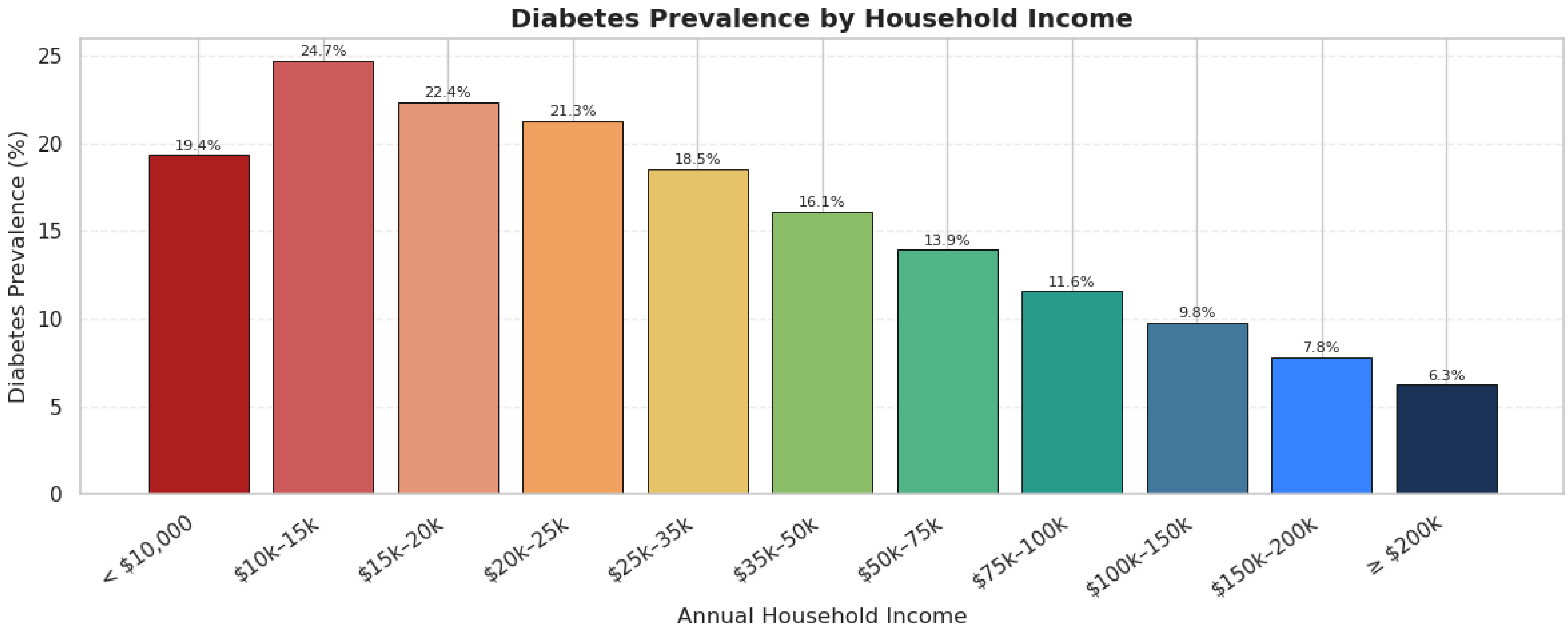
DEMOGRAPHIC RISK FACTORS



- The risk of Type 2 Diabetes generally increases with age due to reduced insulin sensitivity, progressive pancreatic β-cell dysfunction, and the accumulation of metabolic risk factors over time.

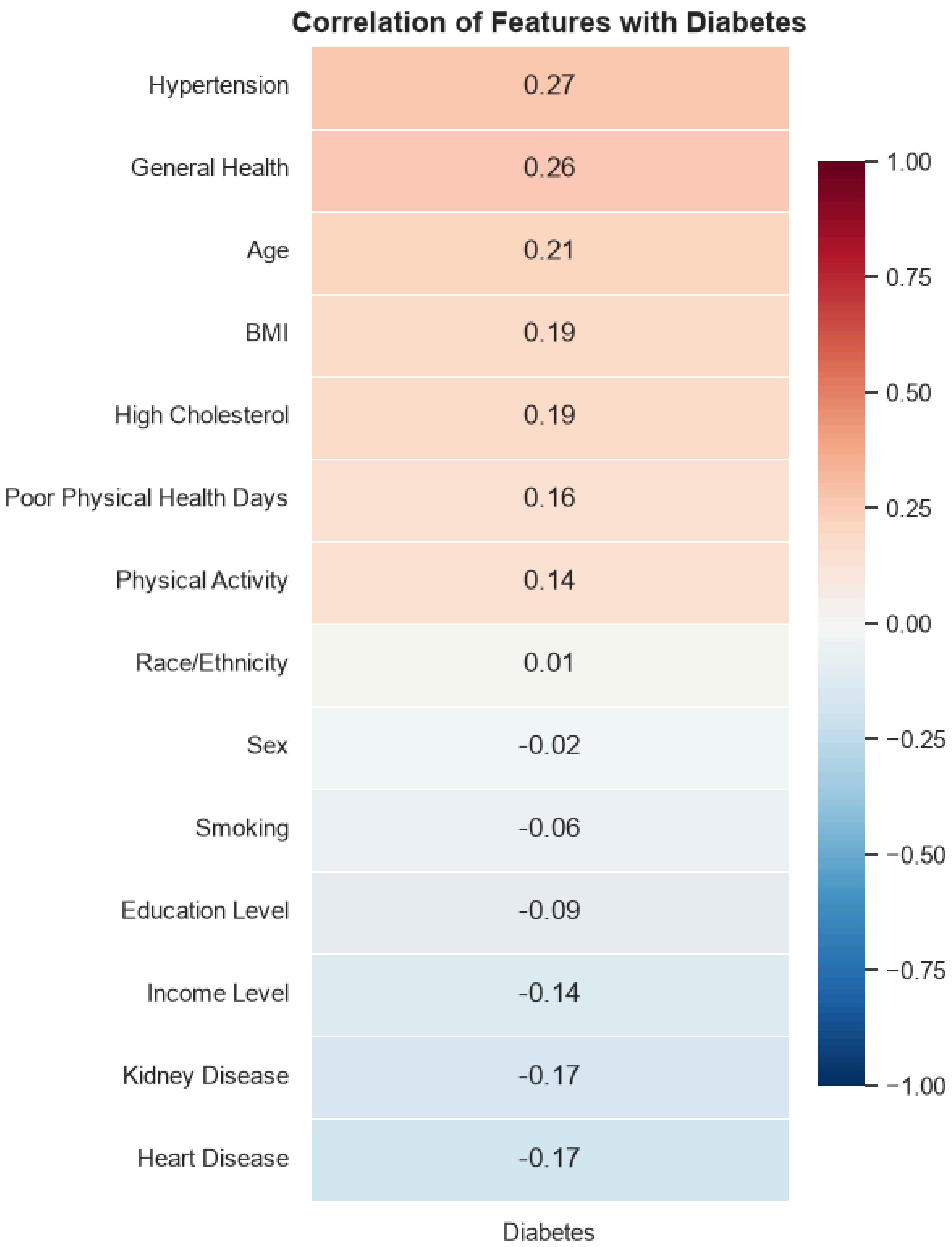


This pattern suggests that higher educational attainment may contribute to improved health literacy, healthier lifestyle choices, better employment opportunities, and greater access to healthcare services thus lowering the prevalence to Diabetes.



This pattern suggests that lower socioeconomic status is associated with a greater burden of Type 2 diabetes, potentially reflecting disparities in healthcare access, nutrition, education, and opportunities for maintaining healthy lifestyles.

CORRELATION OF FEATURES WITH DIABETES



Key Observations

- Age, BMI, general health, and hypertension showed the strongest positive association with diabetes.
- Income and education exhibited weaker negative associations.
- The findings validate the selected predictors for machine learning model development.

Machine Learning Pipeline

End-to-End Workflow for Predicting Type II Diabetes Risk

1. **Dataset Cleaning & Preprocessing**
   Handle missing values, remove outliers, validate types

2. **Feature Engineering**
   Encoding, scaling, new feature creation

3. **Modeling**
   Logistic Regression, Decision tree, Random Forest, XGBoost, LightGBM.

4. **Model Comparison**
   Metric-based ranking

5. **Hyperparameter Tuning**
   GridSearchCV, threshold optimization

6. **Evaluation**
   ROC-AUC, Recall,

7. **Explainability**
   SHAP values, LIME, feature importance

8. **Deployment**
   FastAPI inference backend with a Next.js dashboard

Model Comparison Results

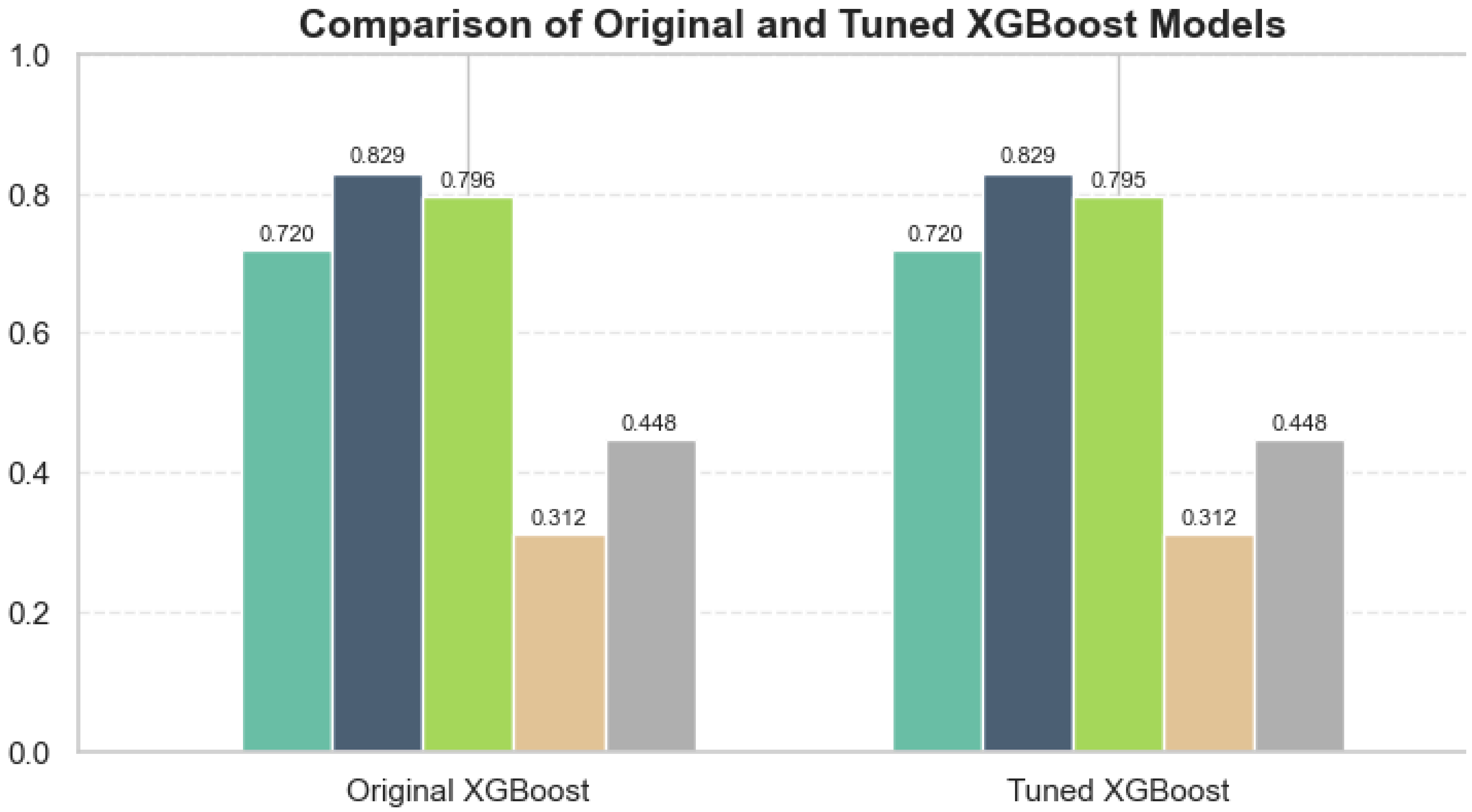
Success Metrics: **Recall** & **ROC-AUC** | Higher = Better at catching diabetic patients

Model	Accuracy	ROC-AUC ★	Recall ★	Precision	F1-Score
XGBoost	0.7198	**0.8291**	0.7961	0.3117	0.4480
LightGBM	0.7157	0.8291	0.7991	0.3087	0.4454
Random Forest	0.7193	0.8239	0.7821	0.3092	0.4432
Logistic Regression	0.7291	0.8222	0.7639	0.3151	0.4462
Decision Tree	0.6803	0.8024	**0.8057**	0.2828	0.4187

🏆 **XGBoost Selected** — Best ROC-AUC (0.829) with strong Recall (0.796)

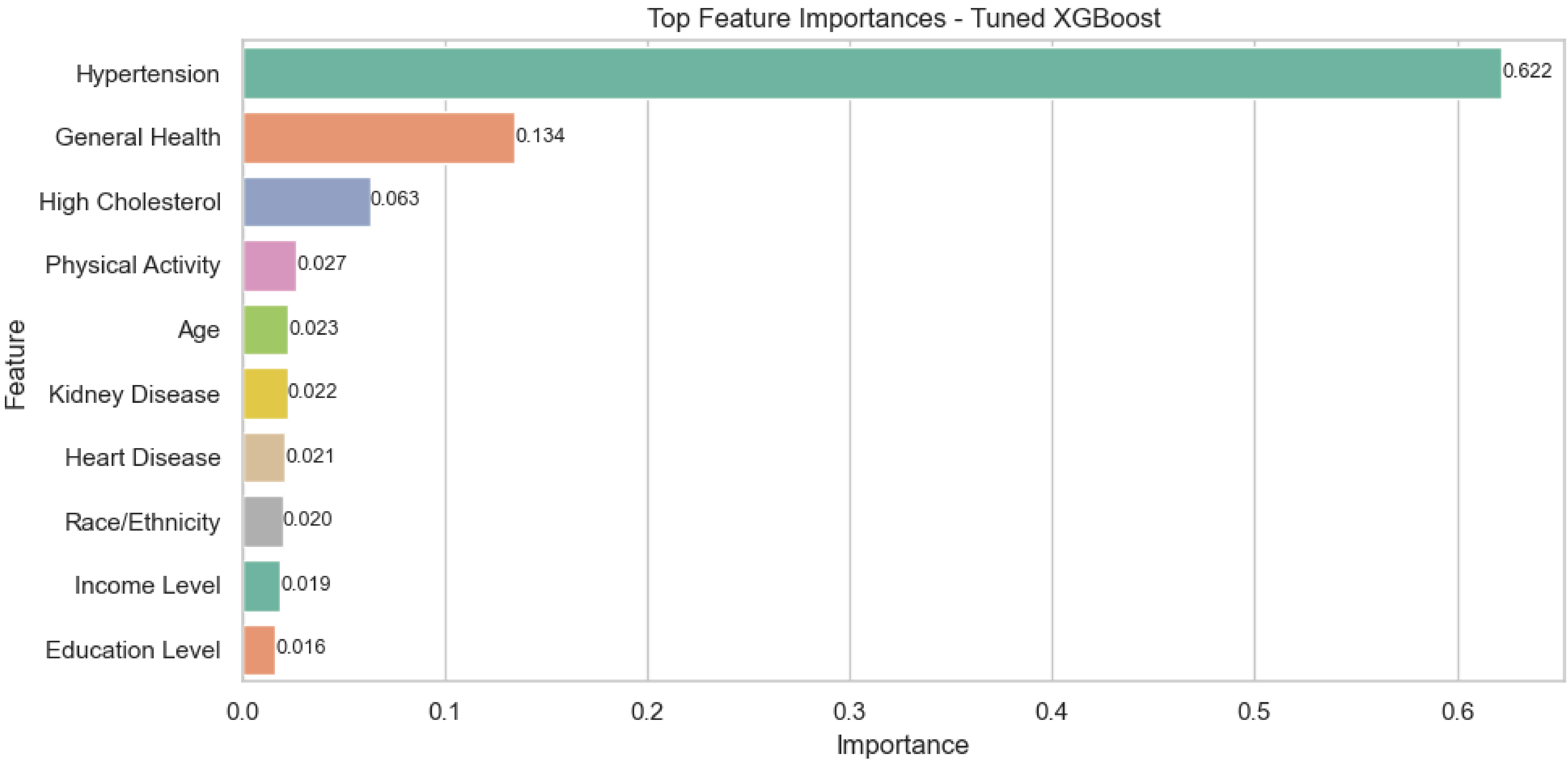
Decision Tree has highest Recall (0.806) but lowest ROC-AUC — XGBoost offers the best balance

Final Model Performance: XGBoost



The original XGBoost model was already well-optimized, and the tuned model did not provide a meaningful performance improvement.

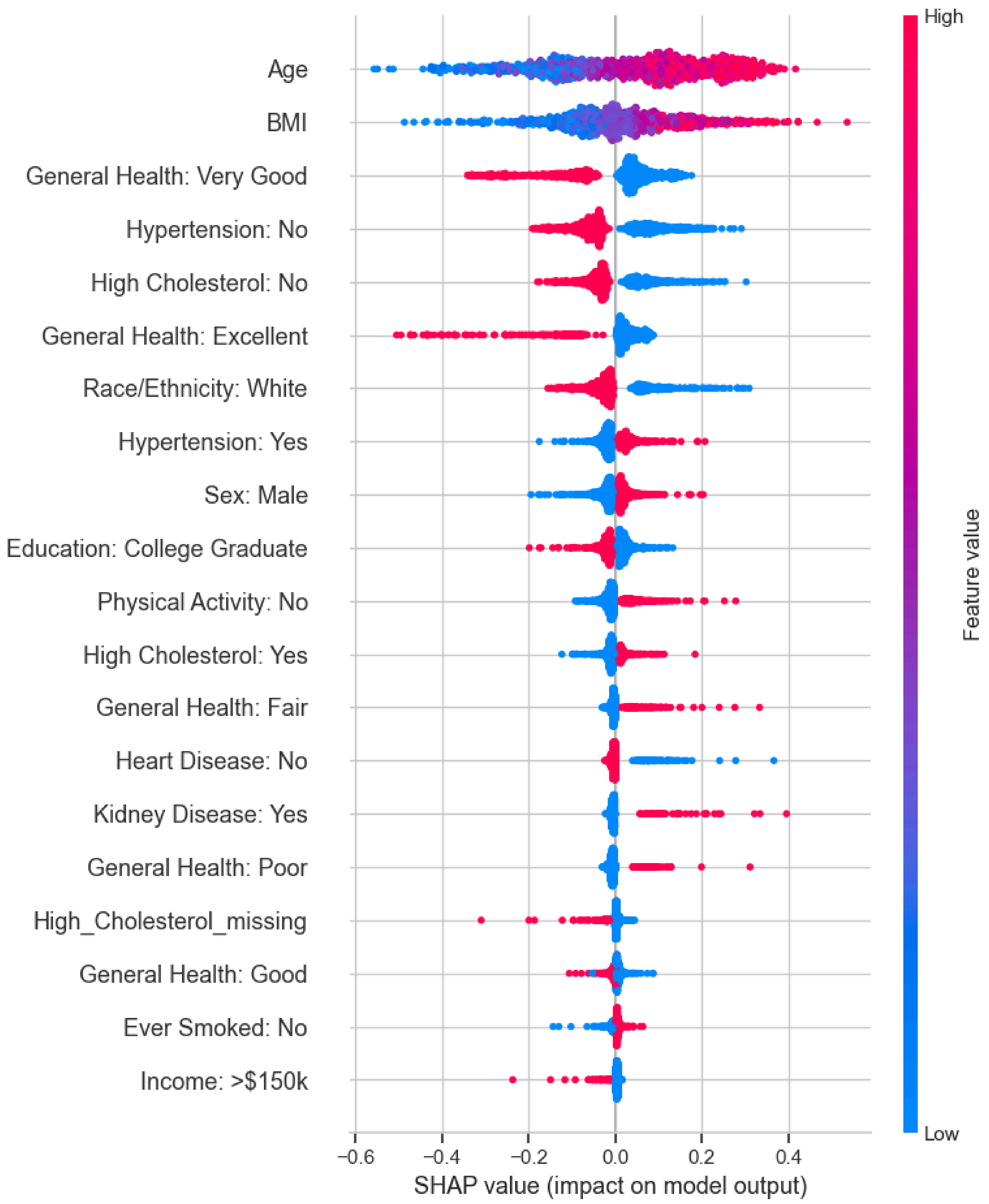
Feature Importance



Top predictors: Hypertension, General Health, High Cholesterol

Additional predictors: BMI, Age, Kidney Disease, Heart Disease, Physical Inactivit

Cardiometabolic factors (cholesterol, BMI, kidney disease, heart disease, age) dominated feature importance.



Model Explainability: SHAP

Top 10 SHAP Features

Feature importance & direction of risk

Feature	Importance	SHAP Insight
Age	Highest	↑ Older age increases risk
Hypertension	High	↑ Hypertension increases risk
BMI	High	↑ Higher BMI increases risk
General Health	High	↑ Poor health increases risk
High Cholesterol	High	↑ High cholesterol increases risk
Physical Activity	Moderate	↓ Activity reduces risk
Education Level	Moderate	↓ Higher edu slightly reduces risk
Income Level	Moderate	↓ Higher income slightly reduces risk
Heart Disease	Moderate	↑ Heart disease increases risk
Kidney Disease	Moderate	↑ Kidney disease increases risk

LIME Explanation — Individual Prediction

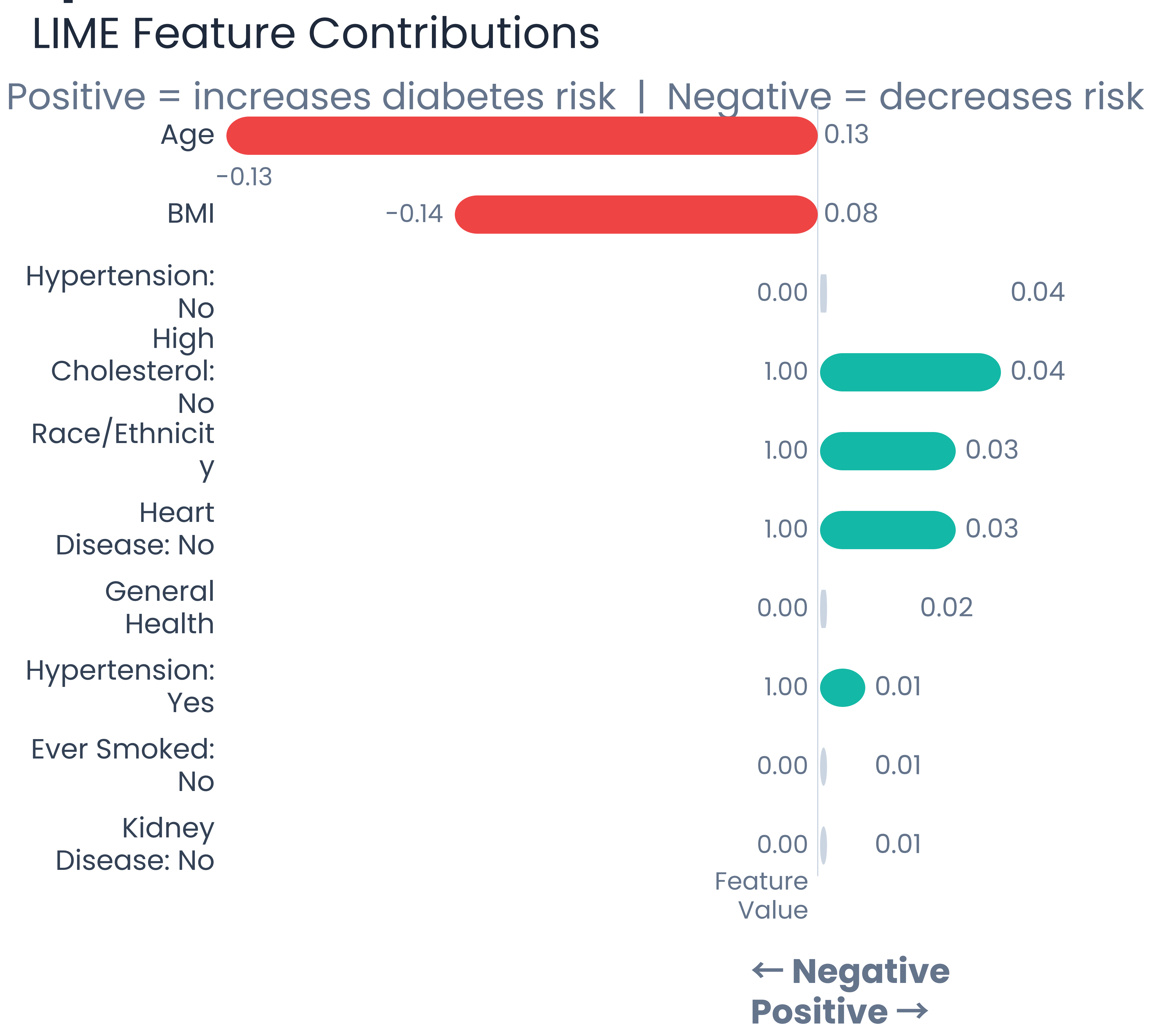
LIME Individual Prediction

68%

Predicted Diabetes Probability

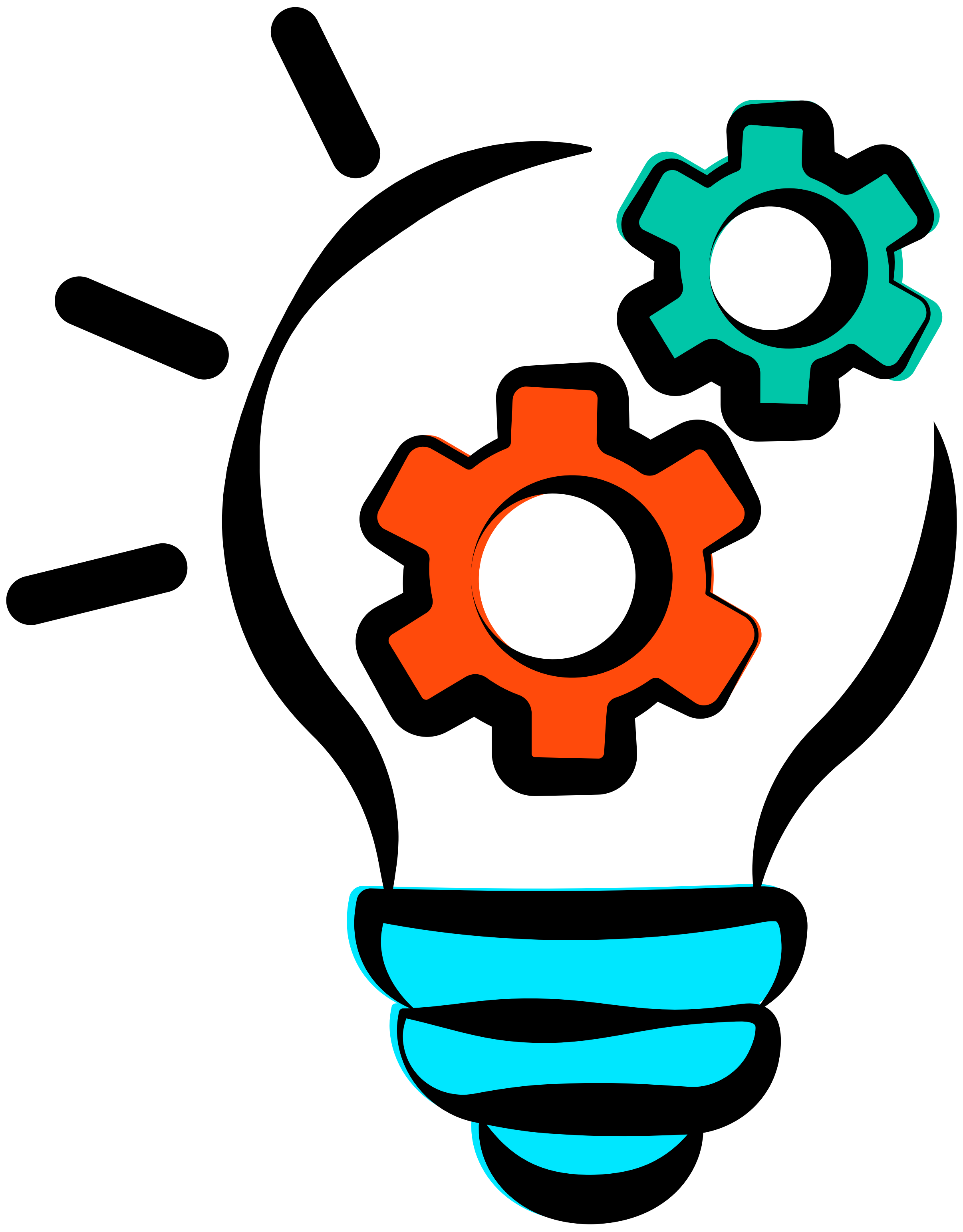
Key Drivers

- Age and BMI are the strongest contributors to the prediction.
- Hypertension, cholesterol, race/ethnicity, and heart disease follow.
- General health, smoking, and kidney disease have smaller effects.



Key findings from the diabetes prediction study

- Age was the strongest predictor of Type 2 Diabetes, with risk increasing among older individuals.
- Hypertension, high BMI, poor general health, and high cholesterol were the key clinical risk factors.
- Physical activity reduced diabetes risk, while higher education and income were associated with slightly lower risk.
- SHAP identified age, hypertension, BMI, general health, and high cholesterol as the most influential features.
- The findings demonstrate that ML can support early identification of individuals at high risk of Type 2 Diabetes.



Limitations & Challenges

Self-reported BRFSS data may contain recall bias and measurement errors

Model generalizability limited to US population; external validation needed

Class imbalance and missing data affect prediction reliability

EHR integration and clinical workflow adoption pose deployment barriers

RECOMMENDATIONS

- Prioritize screening for adults with multiple clinical risk factors
- Use the model as a risk-stratification support tool, not a diagnosis.
- Pair model outputs with intervention pathways.
- Tune the decision threshold to the operational goal.
- Monitor fairness and subgroup performance.

Conclusion

79.61% Recall

82.93% ROC-AUC

~8 in 10 Detected

🎯 Predictive Power

Type 2 Diabetes risk predicted from 14 routine health survey variables without requiring any laboratory tests.

🔍 Explainability

SHAP and LIME transform each prediction into a patient-specific risk profile clinicians can interpret and communicate.

🚀 Deployment

FastAPI + Next.js platform with single/batch prediction, analytics, and PDF reports — ML as a practical clinical tool.

Machine learning, applied responsibly and transparently, has significant potential to improve early detection of chronic diseases and reduce the burden of preventable illness at population level.

Any Question?

Thank You!